

TECHNICAL DOCUMENTATION

Sustainability Dashboard

Data Flows & Business Formulas

Mill Onboarding · Supplier Due Diligence · Traceability
End-to-end mapping across modules and Google Sheets

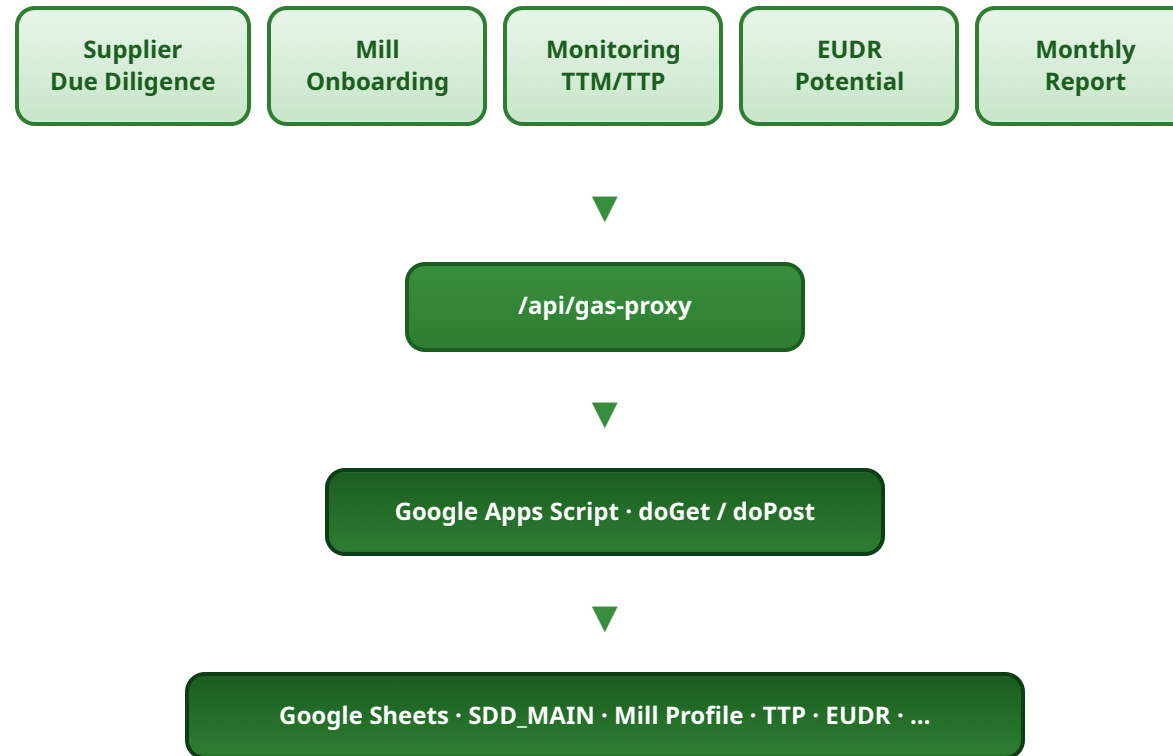
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01 · System Architecture

Frontend → API Proxy → Google Apps Script → Google Sheets



Key principle: Mill scores and risk levels are computed in Google Sheets. The web app never overwrites formula columns.

02 · Supplier Due Diligence (SDD)

Panel #panel-supplier-dd

- **STORAGE** SDD_MAIN · SDD_MILL_LIST · SDD_FFB_LIST
- **KEY** submission_id (SUB-YYYYMMDD-XXXXXX)
- **TYPE** MILL · KCP · TRADER
- Excel upload — Main Form + Traceability A/B
- Screening form — SCR, GRV, PRI fields

- Approve / Hold / Reject (approver decision)
- NBL check vs NBL & Unilever NBL sheets
- Export PDF screening report
- TML / FFB spatial screening (modals)
- Legacy: flat SDD Data sheet (still supported)

03 · SDD Lifecycle

State machine — from import to onboarding complete



04 · SDD Data Destinations

Trigger	Target Module	Backend Function
statusSDD = APPROVED	Contact List Supplier	syncContactFromApprovedSubmission_()
APPROVED + Submitted	Mill Task List	isApprovedSubmittedSddRow_()
Save mill from Task List	Mill Onboarding Profile	add / update sheet mill
mill_ttp_sync payload	Monitoring TTP/TTM	syncTtpFromMillOnboarding_*
NBL check	SCR - No Buy List	runSddNblCheck()
Report period	Monthly Report \$01	fetchSddList()

05 · SDD → Mill Onboarding Mapping

mapSddRowToMillPayload() / mapSddTmlRowToMillPayload()

Mill Column	SDD Source
GROUP NAME	Group Name
COMPANY NAME	Company Name
MILL NAME	Mill Name / KCP Name / TML - Mill Name
COORDINATES	Latitude + Longitude (comma-separated)
MILL CAPACITY (TON/HOUR)	Mill Capacity (Ton/Hour)
SOURCE TYPE	MILL / KCP / TRADER
TRADER NAME / UML ID	Trader parent / TML - UML ID

06 · FFB → TTP/TTM Mapping

One TTP row = one FFB supplier + Main Form context

TTP Column	SDD / Mill Source
GROUP / COMPANY / MILL NAME	Mill Onboarding identity
FFB SUPPLIER GROUP / NAME	FFB - Supplier Group/Name
CATEGORY · LAT/LONG · VILLAGE	FFB location & category fields
CONCESSION / PLANTED AREA	FFB area fields
LEGALITY · ISPO/RSPO/ISCC	FFB certification
FFB SUPPLY to MILL (TON)	FFB - Total Supply FFB (Ton)

Full reference: [docs/sdd-to-ttm-ttp-field-mapping.md](#)

07 · Mill Onboarding — Entry Points

Mill Profile (CPO/PK) · Mill Waste (POME/SHELL)

- **Manual Add/Edit**

Modal #btn-add-mill

- **SDD Task List**

"+ Add to Mill" from approved screening

- **Supply Excel Import**

Upload → draft → submit

- CPO/PK → **mill** sheet

- POME/SHELL → **millWaste** sheet

- KCP plant → **FACILITY NAME PK**

- Refinery → **FACILITY NAME CPO**

- Staging: **Supply Import Draft**

08 · Mill Onboarding → Outputs

Consumer	Join Key	Data Used
Monitoring TTP/TTM	GROUP COMPANY MILL UML	Identity + FFB sync
Monthly Report §02	Period + risk level	High-risk MILL rows
Facility Performance	FACILITY NAME CPO/PK	Supply qty, traceability %
Questionnaire Monitoring	group company mill	Mill registry
EUDR Potential	group company mill	Identity sync
Mill Executive Report	Quarter	Supply, NBL, certification

09 · Traceability Data Overview

Panel #panel-ttm-ttp — Year filter drives all KPI cards

	CPO	PK
TTM Traceable Mill Onboarding sheet	Coordinate-based supply %	Coordinate-based supply %
TTP Traceable Monitoring TTP/TTM sheet	FFB → CPO traceability %	FFB → PK traceability %

● **TTM data source:** Mill Onboarding Profile rows (selected year, all quarters)

● **TTP data source:** Monitoring TTP/TTM rows (filtered by year)

● **UI function:** renderTtpTraceableStats_()

● Also feeds: Monthly Report KPIs (mrdBuildTraceTotalsForReport_)

● Also feeds: Facility Performance weighted %

● Also feeds: Mill Executive Report traceability charts

● Category mix bar: supplier type % from FFB supply volume

10 · TTM Traceable % — Formula

ttpCalcTtmCoordinatePct() · Supply-weighted coordinate check

CPO TTM %

Pool: SOURCE TYPE = MILL / TRADER / REFINERY (never KCP)
PRODUCT SUPPLY contains CPO — or blank with SUPPLY CPO > 0
Traceable row: valid lat/lng coordinate
Formula: $\Sigma(\text{SUPPLY CPO traceable}) \div \Sigma(\text{SUPPLY CPO total}) \times 100$

PK TTM %

Pool: same SOURCE TYPE filter
SUPPLY PK column > 0 (full-year total, not per quarter)
Traceable row: valid lat/lng coordinate
Formula: $\Sigma(\text{SUPPLY PK traceable}) \div \Sigma(\text{SUPPLY PK total}) \times 100$

Per-mill shortcut (Monthly Report): mrdMillTtmRowPct() → 100% if valid coordinate, else 0% (binary per row).

11 · TTP Traceable % — Formula

ttpAggregateTotalTraceablePct_() · Matches sheet footer totals

Product	Numerator (Σ)	Denominator (Σ)	Display
CPO	CPO TRACEABLE VOLUME	CPO SUPPLY TO REFINERY	#ttp-stat-cpo-traceable
PK	PK TRACEABLE VOLUME	PK SUPPLY TO KCP	#ttp-stat-pk-traceable

- **Primary method:** SUM(numerator) ÷ SUM(denominator) × 100 — same as Google Sheet footer
- **Fallback:** if volume columns missing → average of % **CPO TRACEABLE** or % **PK TRACEABLE** per row
- **Format:** 2 decimal places (e.g. 94.13%) via ttpFormatTraceablePct_()
- **Row filter:** ttpIsDataRow_() excludes header echoes and "Total Traceable" footer rows

12 · TTP Sheet — Key Columns & Conversions

Per FFB supplier row in Monitoring TTP/TTM registry

- **Input:** FFB SUPPLY to MILL (TON)
 - **Conversion:** FFB → CPO (20%) · FFB → PK (5%)
 - **Output supply:** CPO SUPPLY TO REFINERY · PK SUPPLY TO KCP
 - **Traceable volume:** CPO/PK TRACEABLE VOLUME
 - **Per-row %:** % CPO TRACEABLE · % PK TRACEABLE
- **Sheet totals:** Total CPO % Traceable · Total PK % Traceable
 - **Identity:** GROUP / COMPANY / MILL / UML ID
 - **FFB context:** SUPPLIER GROUP/NAME · CATEGORY · LAT/LONG
 - **Sync metadata:** submission_id · ffb_line_id · synced_at
 - **Monitoring preserved on re-sync:** MSD · trace volumes · % columns

13 · Category Mix & Group Averages

Supplier type breakdown on Traceability panel

Category Mix (TTP panel bar)

Function: ttpSupplierPctByCategory_
Weight = FFB SUPPLY to MILL (fallback weight = 1 per row)
% per bucket = bucket supply ÷ total supply × 100
Buckets: Owned Estate · External Estate · Dealer · Plasma · Cooperative

Registry Group Average

Function: ttpGroupAvgPctHtml_
Uses same ttpAggregateTotalTraceablePct_
Shows CPO/PK aggregate % in grouped table rows

Category record count mix: ttpComputeCategoryMix_
— % of rows by supplier category (largest-remainder rounding, labels sum to 100%).

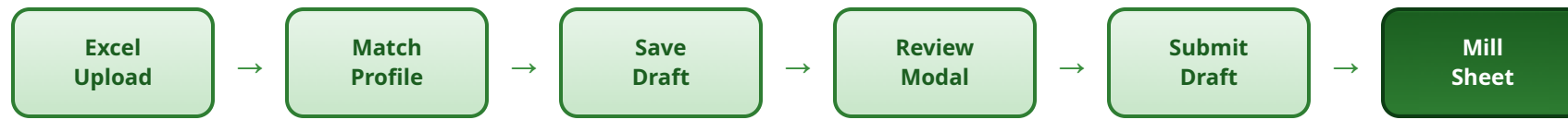
14 · Monthly Report & Facility Performance

Downstream consumers of TTM/TTP formulas

Module	Function	Formula
Monthly Report KPI	mrdBuildTraceTotalsForReport_()	TTM + TTP aggregate for report year (TTP year = report year – 1)
Per-mill report row	mrdBuildTraceRowsForReport_()	Joins mill ↔ TTP by company; avg % CPO/PK TRACEABLE
Facility TTP %	pfBuildRows_() / pfBuildPkGroups_()	$\Sigma(\text{supply qty} \times \text{TTP}\%) \div \Sigma(\text{supply qty}) \times 100$
Facility TTM %	pfCalcTtmFacilityTrace_()	$\Sigma(\text{supply qty} \times \text{TTM}\%) \div \Sigma(\text{supply qty}) \cdot \text{TTM} = 100 \text{ or } 0 \text{ per seller}$
ISPO supply %	pfCalcIspoSupplyPct_()	$\Sigma(\text{ISPO-certified qty}) \div \text{total supply} \times 100$
Untraceable mills	Monthly Report §03	Mills with no FFB supplier in TTP for trace year

15 · Supply Task List

Embedded in the Mill Onboarding panel



- match_status: **matched** · **new** · group_mismatch
- Formula columns restored from row above — never overwritten
- Repair drift: **reconcileSupplyDraft_()**

16 · Google Sheet Formulas (Read-Only)

MILL_FORMULA_HEADERS_ — never written from the web app

● SCORE · TOTAL SCORE · LEGALITY SCORE

● DEFORESTATION / BURN / PEAT SCORE

● TOTAL GRIEVANCES · POLICY · CERTIFICATION

● RISK LEVEL · RESULT RISK LEVEL

● BUYER NO BUY LIST · SUPPLIER STATUS

● PERCENTAGE SUPPLY CPO/PK/ISCC/INS/SHELL

● PRODUCT SUPPLY · STATUS SUPPLY CPO/PK

● TOTAL POME SUPPLY · MAX/REMAINING STOCK

● TOTAL SCORE SUPPLY · TOTAL RISK LEVEL

● SD Monitoring: DAY LEFT · Status · Result

17 · Other App Formulas

Grievance · EUDR · Mill risk — computed in frontend

Grievance Risk Score

Total = sum of 6 indicators (score 1–3)
≤ 8 → Low · ≤ 11 → Medium · > 11 → High
src/grievance-risk.js

EUDR Potential Status

All enabled criteria pass → Potential
Any fail → Not Potential
FFB % by TTP category vs threshold (default 70%)
eudrComputeStatus_()

Mill Risk Reason

UI pills from field gaps: coordinate, legality, NDPE, cert, deforestation, NBL
Uses RESULT RISK LEVEL from sheet
src/mill-risk-reason.js

Mill Executive Report

Risk buckets, NBL counts, supply by province/facility
Traceability pie: traceable vs untraceable supply
src/mill-executive-report.js

18 · Grievance & Mill Risk Reason

● **Grievance** — all 6 indicators required before classification

- Publish · Subject · Repeat · Consequence
- Group Scale · No Response

● **Mill Risk Reason** — UI pills derived from field gaps

- No Coordinate · Legality Not Complete
- Non APL · No NDPE · No Certification
- Deforestation > 0 ha · On No Buy List

19 · Full Module Connection Map

How data flows across the entire dashboard



20 · Google Sheets Registry

Key	Tab Name	Module
mill / millWaste	Mill Onboarding Profile / Waste	Mill Onboarding
sddMain/Mill/Ffb	SDD_MAIN / MILL_LIST / FFB_LIST	Supplier DD
ttp	Monitoring TTP/TTM	Traceability
grievance	Grievance Monitoring	Grievance
eudrPotential	EUDR Potential	EUDR
contactSupplier	Contact List Supplier	Contact List

21 · SDD vs EUDR DDS

Two separate modules — no direct coupling

Aspect	SDD (Supplier Due Diligence)	EUDR DDS
Panel	#panel-supplier-dd	#panel-due-diligence-statement
Storage	SDD_MAIN + child sheets	EUDR DDS + child sheets
Purpose	Internal screening + onboarding	EU EUDR export package
Output	Task List → Mill → TTP	DOCX/PDF for buyers

22 · Validation Rules

Guard rules that prevent data inconsistency

- Submitted SDD cannot be downgraded to Draft
- Task List: SCR = submitted AND statusSDD = APPROVED
- TTP sync: full identity (GROUP/COMPANY/MILL/UML)
- Supply submit: match_status = matched or new
- Mill save: formula columns stripped before write
- EUDR: all enabled criteria must pass
- Grievance: all 6 indicators required before classification
- Coordinates: normalizeCoordinate() + recoverCoord()

23 · Code Reference

Area	File / Function
Backend CRUD + TTP sync	scripts/GoogleAppsScript-backend-v3-full.gs
TTM / TTP formulas	src/main.js — ttpCalcTtmCoordinatePct_ · ttpAggregateTotalTraceablePct_
Traceability UI	partials/panel-ttm-ttp.html · renderTtpTraceableStats_()
Monthly report trace	src/monthly-report-ui.js — mrdBuildTraceTotalsForReport_()
Facility performance	src/main.js — pfCalcTtmFacilityTrace_ · pfBuildRows_()
TTP field mapping doc	docs/sdd-to-ttm-ttp-field-mapping.md

Thank You

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